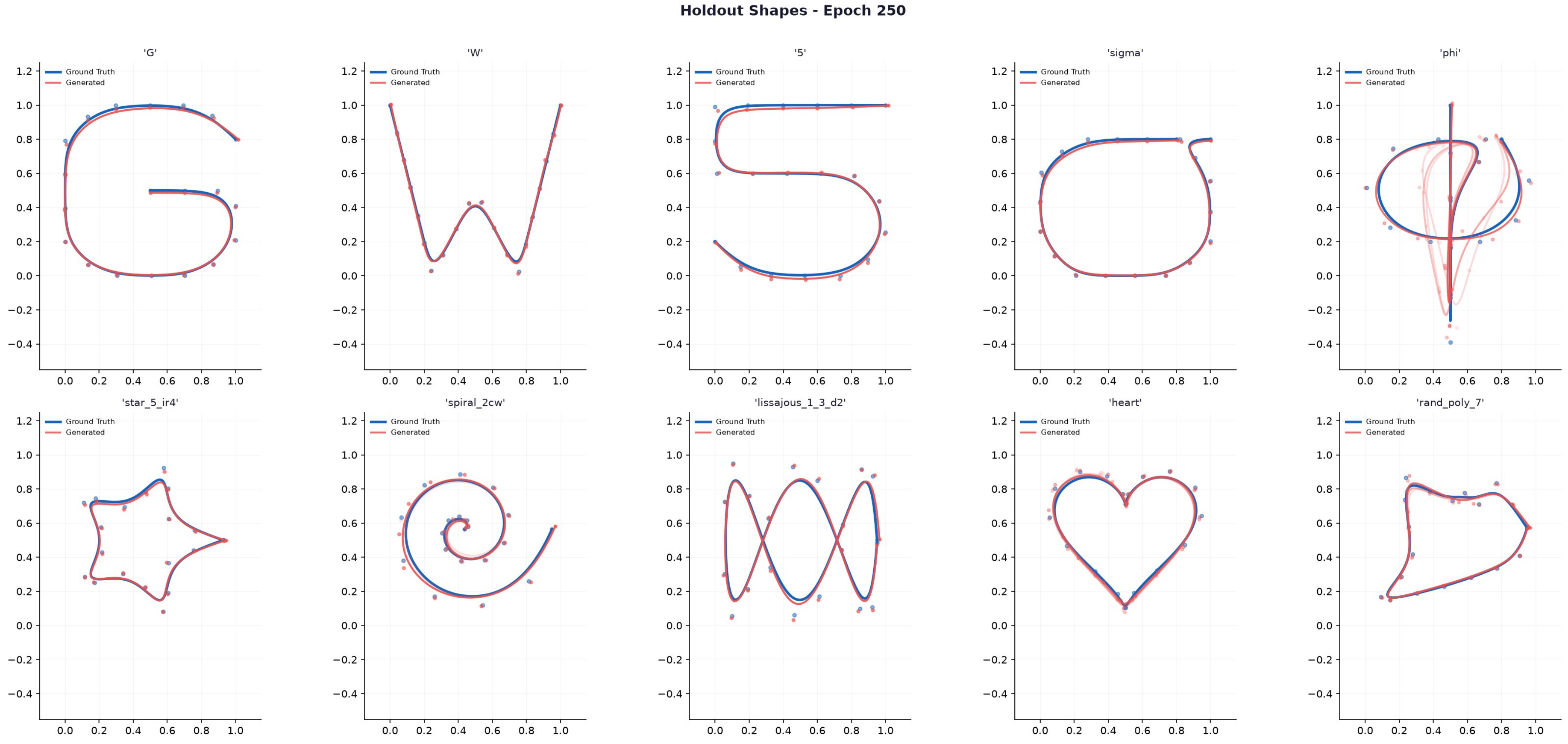
Meeting vom 14.08.

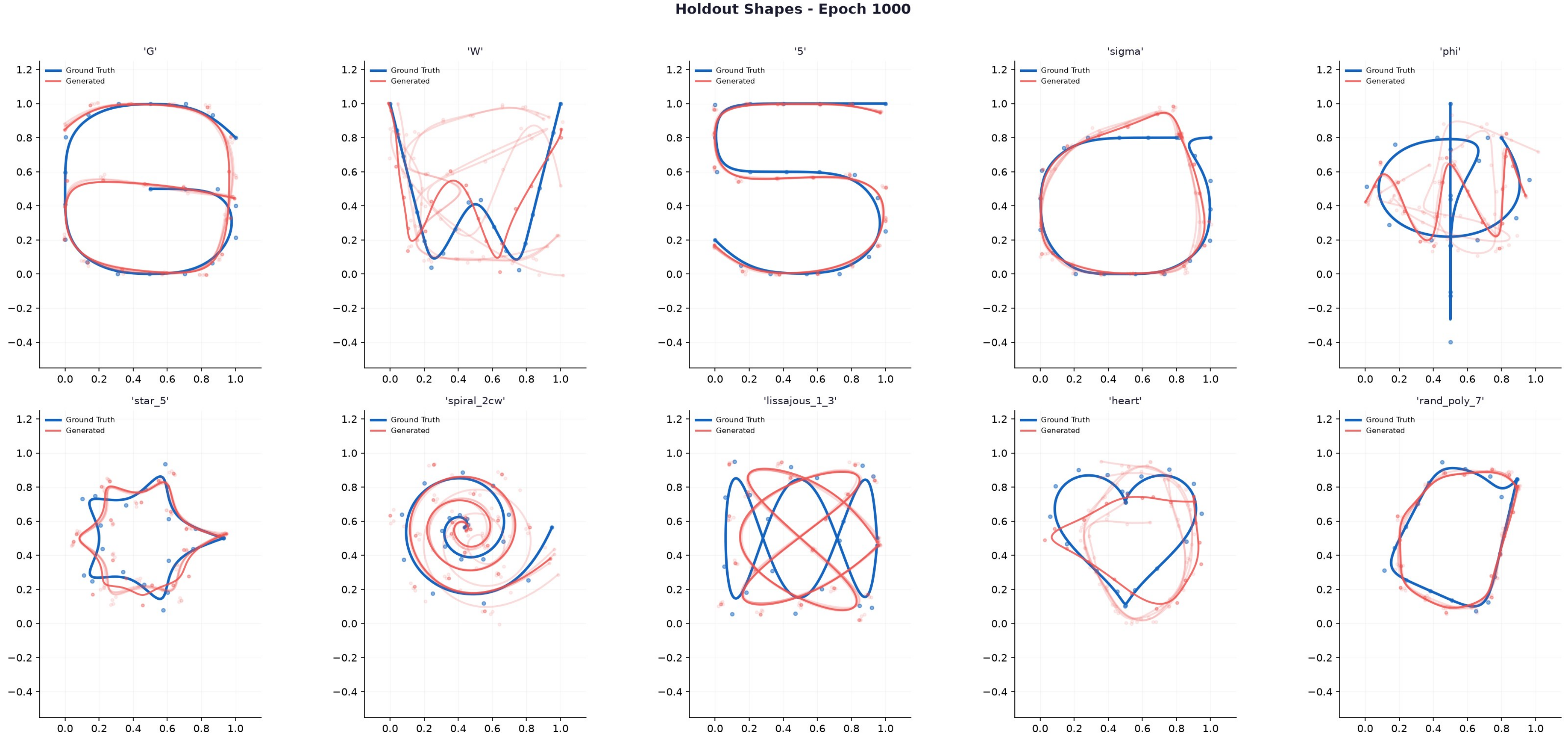
Formrekonstruktion

- Conditioning with crossattention blocks in the U-net
- One version with direct tokenization on the shape waypoints with convolution layers
- Second version with spectral fourrier projection and tokenization with spectral coefficients

Waypoints


Holdout Shapes - Epoch 250
'G'
'W'
'5'
'sigma'
'phi'
'star_5_ir4'
'spiral_2cw'
'lissajous_1_3_d2'
'heart'
'rand_poly_7'
Ground Truth
Generated
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
0.0
0.2
0.4
0.6
0.8
1.0

Spectral



Holdout Shapes - Epoch 1000
'G'
'W'
'5'
'sigma'
'phi'
'star_5'
'spiral_2cw'
'lissajous_1_3'
'heart'
'rand_poly_7'
Ground Truth
Generated

Tested ergodic coverage learning

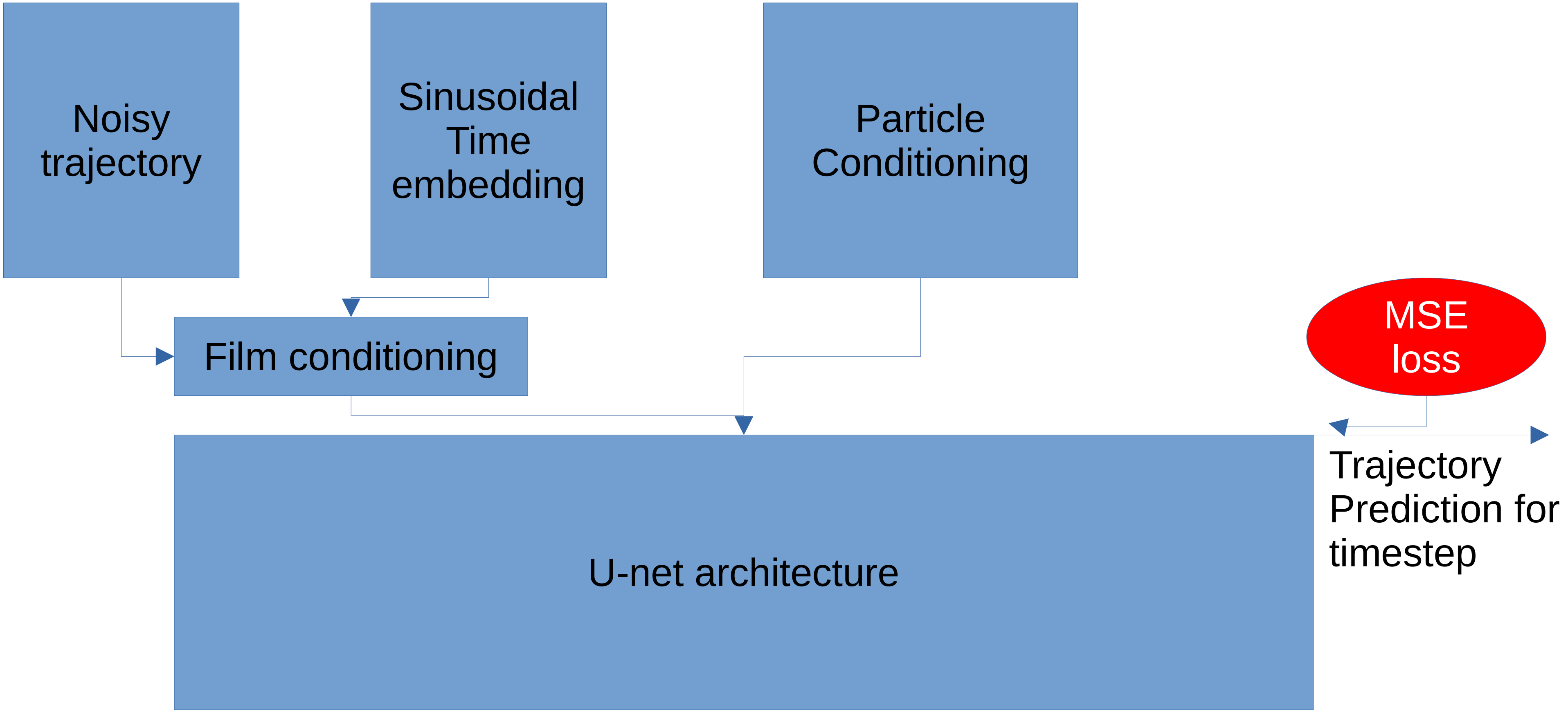
- Dataset created: Generated 750 shapes
  - Random shapes, shapes based on numbers, (asian) letters etc.
  - Sum of gaussian distributions either as lines with gaussian decay or gaussians with defiened covaraınces and averages put together
- Target trajectory:
  - Connect the high probability areas. Define wavy pattern along covariances
  - SVGD solver applied to heuristical initialization

Target Distributions & Init Trajectories (Batch 1)
A
greek_upper_0
korean_5
digit_5
a_lc
rand_gmm_10
rand_gmm_20
rand_ana_poly_10
rand_ana_poly_20
rand_ana_poly_30
rand_gmm_complex_0
rand_gmm_complex_10
rand_gmm_complex_20
rand_gmm_complex_30
rand_ana_poly_complex_0
rand_ana_poly_complex_10
rand_ana_poly_complex_20
rand_ana_poly_complex_30
organic_0
organic_10
organic_20
organic_30
cjk_0
cjk_10
cjk_20
rand_ana_poly_26
rand_ana_poly_62
rand_ana_poly_56
rand_ana_poly_161
rand_ana_poly_99
rand_gmm_complex_74
rand_ana_poly_173
rand_ana_poly_106
O
m_lc
digit_9
organic_18
rand_gmm_85
rand_gmm_19
greek_lower_4
rand_ana_poly_complex_56
rand_ana_poly_90
rand_gmm_159
rand_gmm_108
rand_ana_poly_137
rand_ana_poly_complex_58
rand_gmm_complex_21
rand_ana_poly_complex_21
cjk_16
rand_ana_poly_131
rand_ana_poly_111
rand_gmm_73
rand_gmm_36
rand_ana_poly_133
rand_ana_poly_complex_51
cjk_48
rand_ana_poly_complex_45
rand_ana_poly_complex_6
rand_gmm_13
rand_ana_poly_2
K
p_lc
cjk_45
Q
rand_ana_poly_complex_53
rand_gmm_44
rand_gmm_complex_49
rand_gmm_140
greek_upper_6
rand_ana_poly_1
rand_gmm_71
cjk_58
rand_ana_poly_127
rand_gmm_155
rand_gmm_3
rand_ana_poly_28
G
greek_upper_20
rand_ana_poly_5
organic_5
greek_lower_10
rand_gmm_107
Z
rand_ana_poly_141
rand_gmm_154
rand_ana_poly_9
organic_1
rand_gmm_complex_7
rand_gmm_complex_55
rand_gmm_complex_46
greek_lower_19
j_lc
cjk_53
cjk_60
rand_gmm_134
S
rand_ana_poly_166
L
rand_ana_poly_89
rand_gmm_1

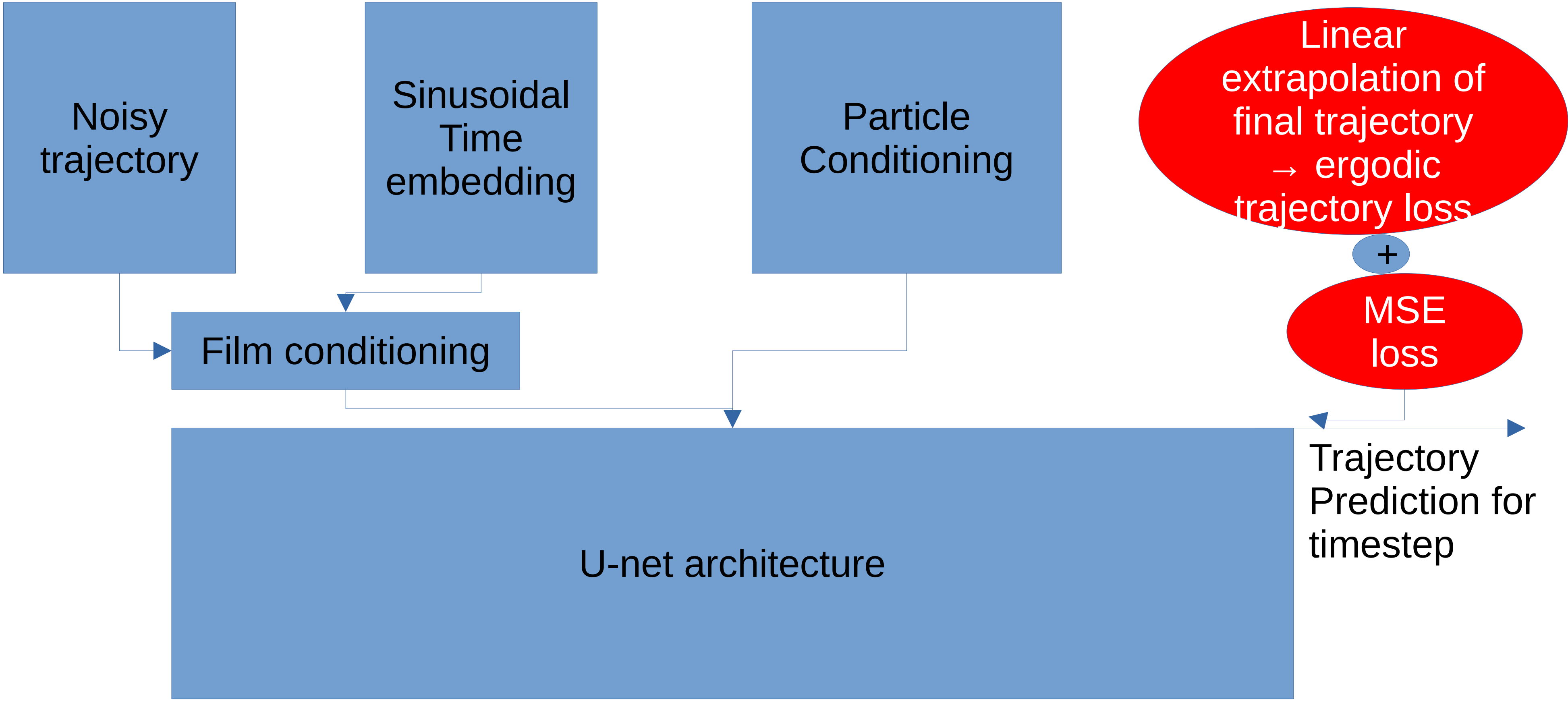
Ergodic coverage flow matching

- Conditioning based on sampling particles proportional to the target distribution
- Also tried spectral decomposition but refuted due to better pointwise conditioning

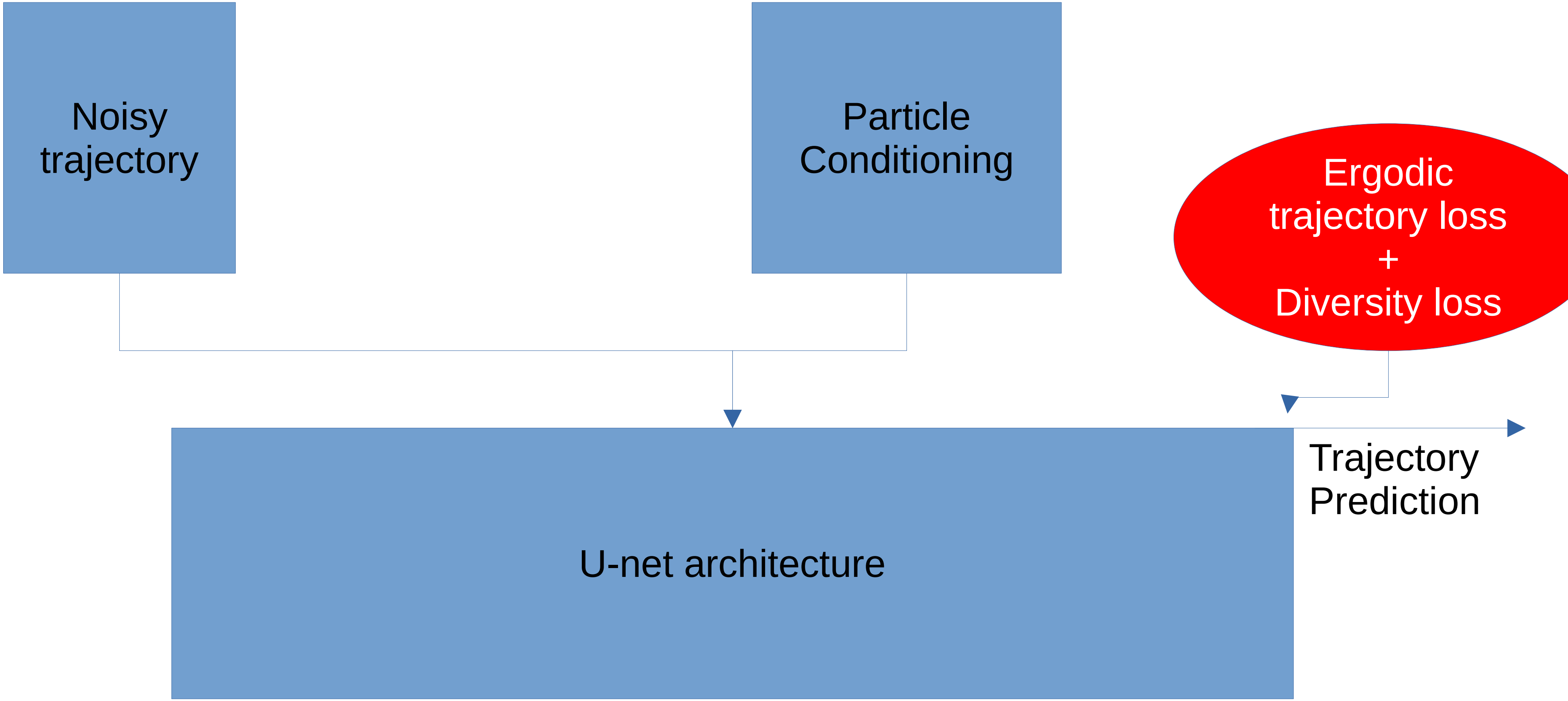
Architecture flow matching



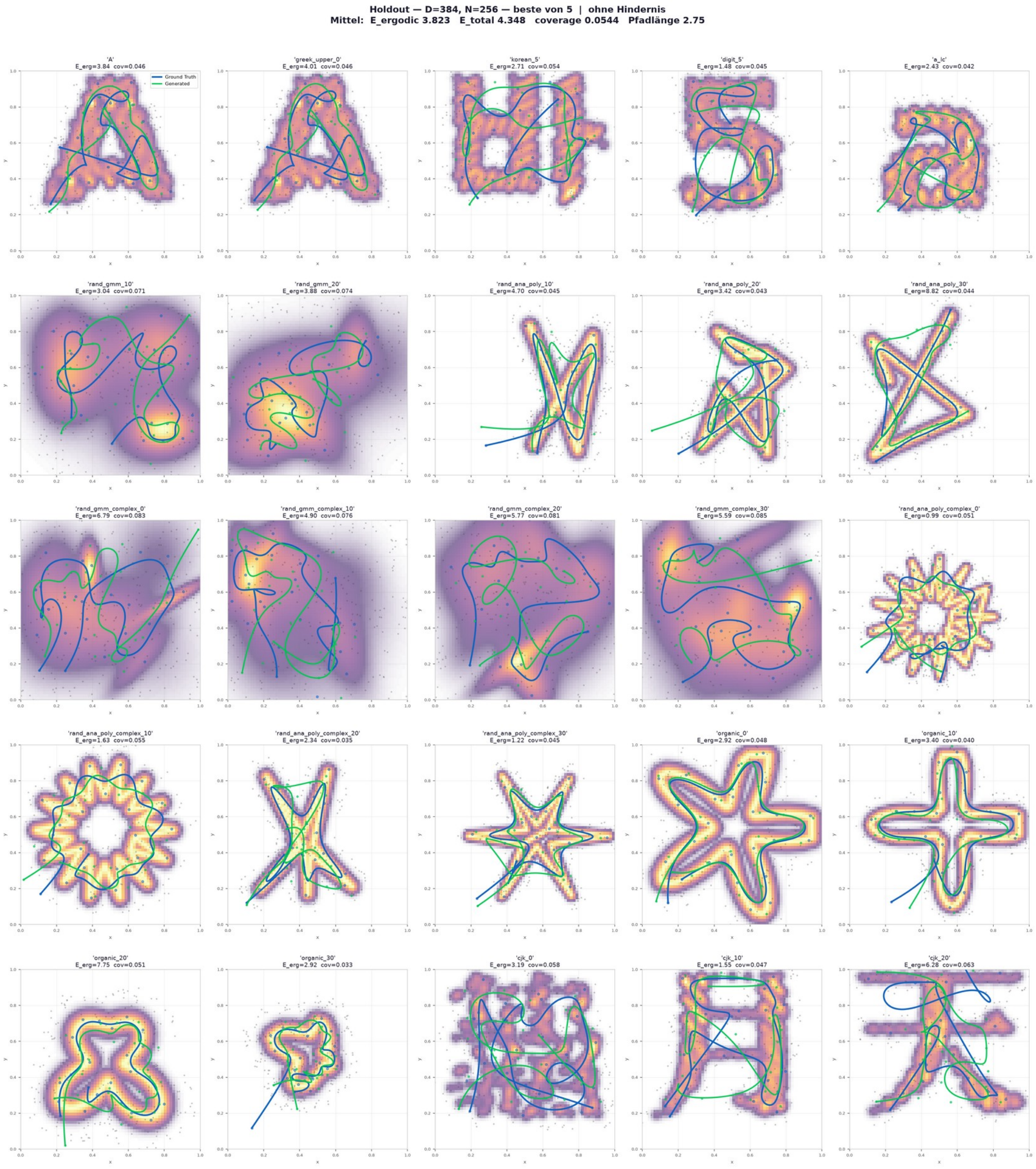
Architecture flow matching + ergodic loss



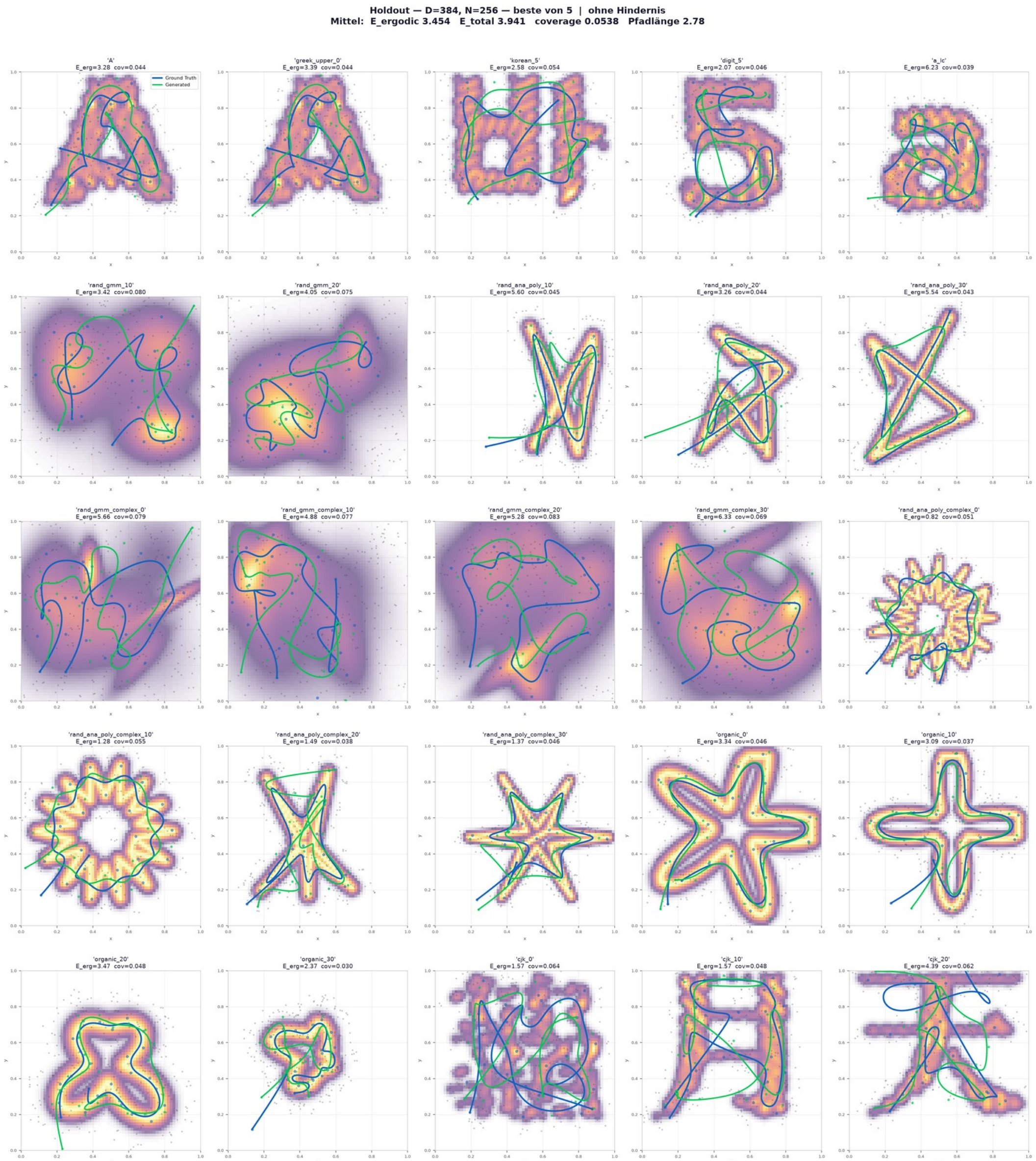
Architecture self supervised



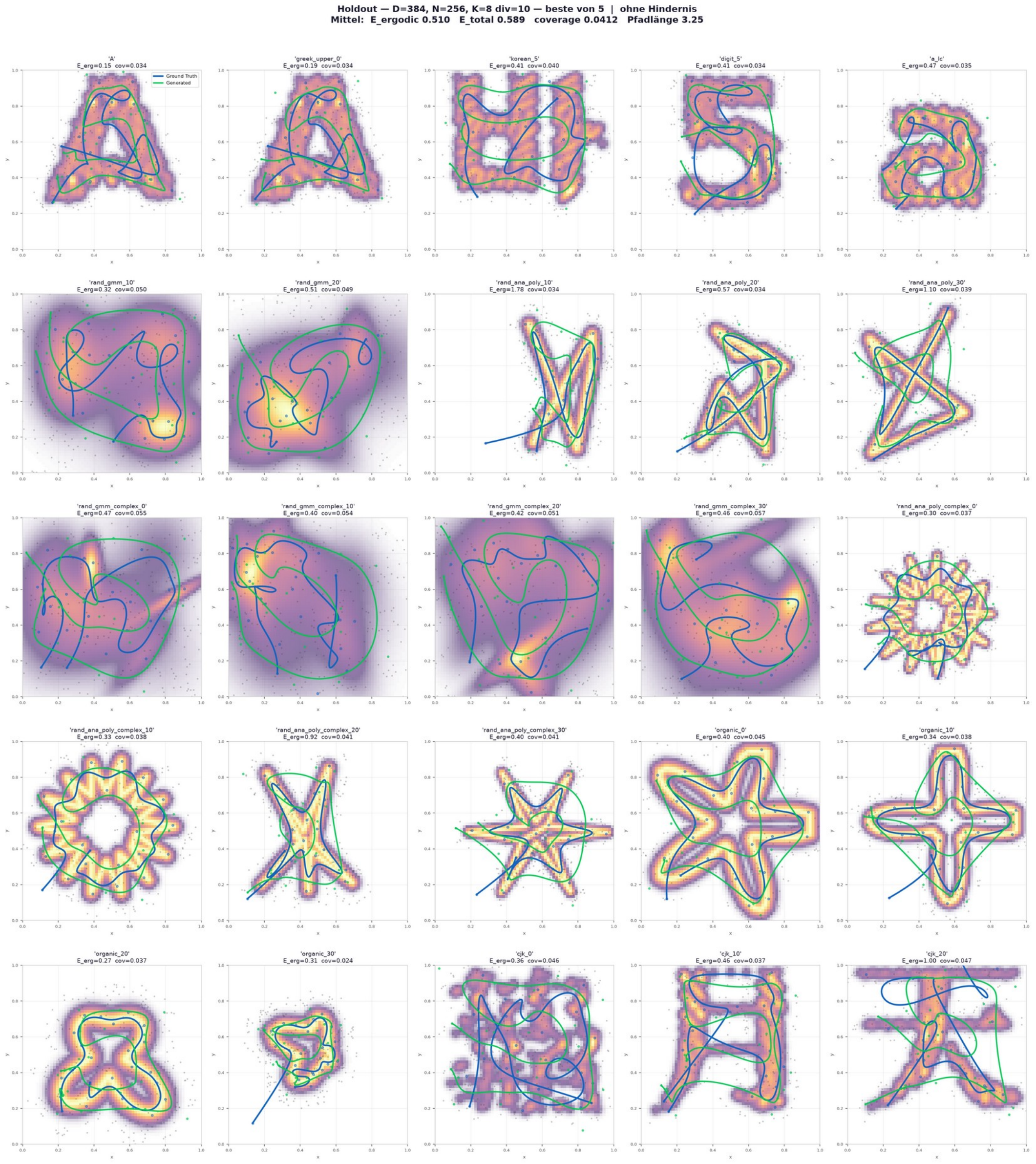
Results: flow matching



Results: flow matching + ergodic loss

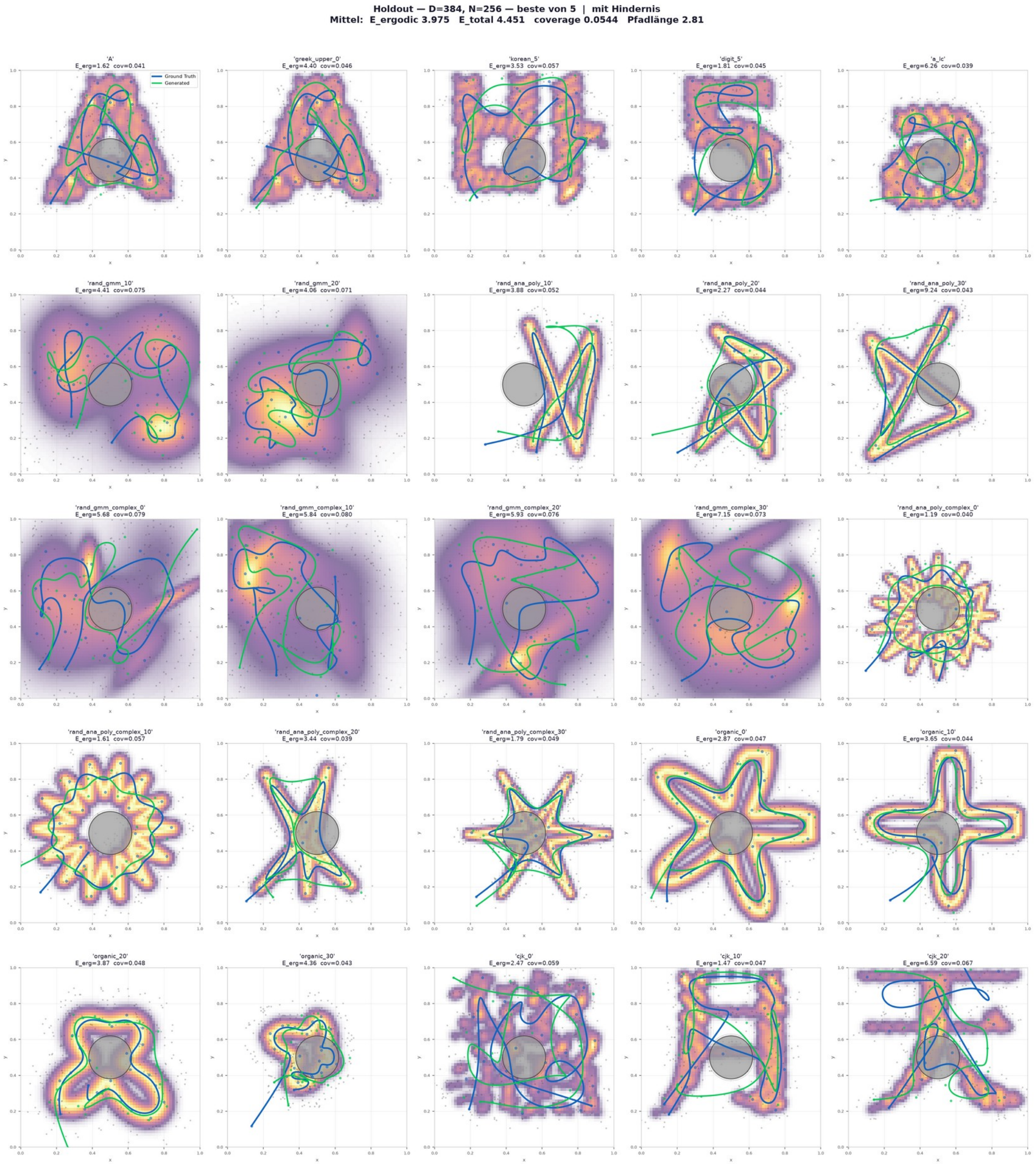


Results: self supervised



Obstacle inferenc

- Calculate rupulsion force during each eul step during inference



Questions/Plans for next week